

Measurement incompatibility and genuine multipartite steering

Lucas E. A. Porto*, Lucas Tendick, Daniel Cavalcanti, Roope Uola, Marco Túlio Quintino*

*Sorbonne Université, CNRS, LIP6, Paris, France

arXiv:2603.25345

1. Introduction

- Measurement incompatibility and bipartite quantum steering display a strong connection: every set of incompatible measurements leads to bipartite steering.
- Despite such a close link between these concepts in bipartite scenarios, little is known in the multipartite setting, where notions of *genuine* multipartite correlations play major roles.
- **Question:** Can every set of incompatible measurements produce genuine multipartite steering?

2. Preliminaries

2.1. Bipartite steering

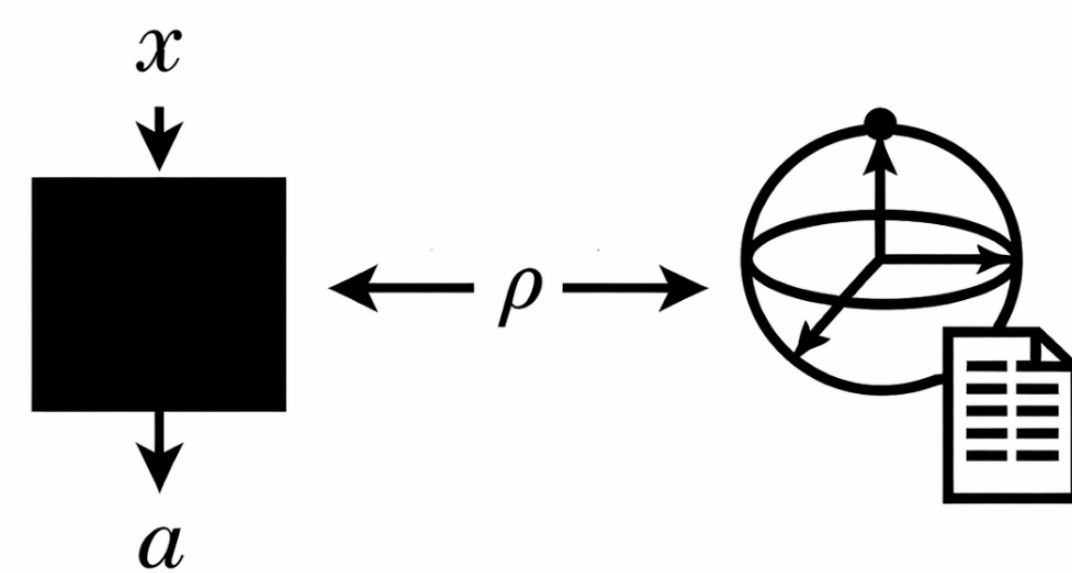


Figure 1: Scheme of a bipartite steering scenario.

Steering assemblage: $\sigma_{a|x}^B = \text{Tr}_A(A_{a|x} \otimes \mathbb{1} \rho^{AB})$

A steering assemblage $\{\sigma_{a|x}\}_{a,x}$ *does not demonstrate steering* if there is a set of states $\{\rho_\lambda\}_\lambda$ and probability distributions $p(\lambda)$ and $p(a|x, \lambda)$ such that

$$\sigma_{a|x} = \sum_{\lambda} p(\lambda) p(a|x, \lambda) \rho_\lambda$$

2.2 Joint measurability and its connection to steering

A set of measurements $\{A_{a|x}\}_{a,x}$, with $A_{a|x} \geq 0$ and $\sum_a A_{a|x} = \mathbb{1}$ is *jointly measurable* if there exists a single measurement $\{G_\lambda\}_\lambda$, with $G_\lambda \geq 0$ and $\sum_\lambda G_\lambda = \mathbb{1}$, and probability distributions $p(a|x, \lambda)$ such that

$$A_{a|x} = \sum_{\lambda} p(a|x, \lambda) G_\lambda.$$

Otherwise, the measurements $\{A_{a|x}\}_{a,x}$ are *incompatible*.

In Refs.[1, 2], the authors prove that

Theorem

$\{A_{a|x}\}_{a,x}$ is incompatible $\Leftrightarrow \text{Tr}_A(A_{a|x} \otimes \mathbb{1} |\Phi^+\rangle\langle\Phi^+|)$ exhibits steering

That is, every set of incompatible measurements leads to bipartite steering.

3. Genuine multipartite steering (GMS)

3.1. One-sided device-independent scenarios

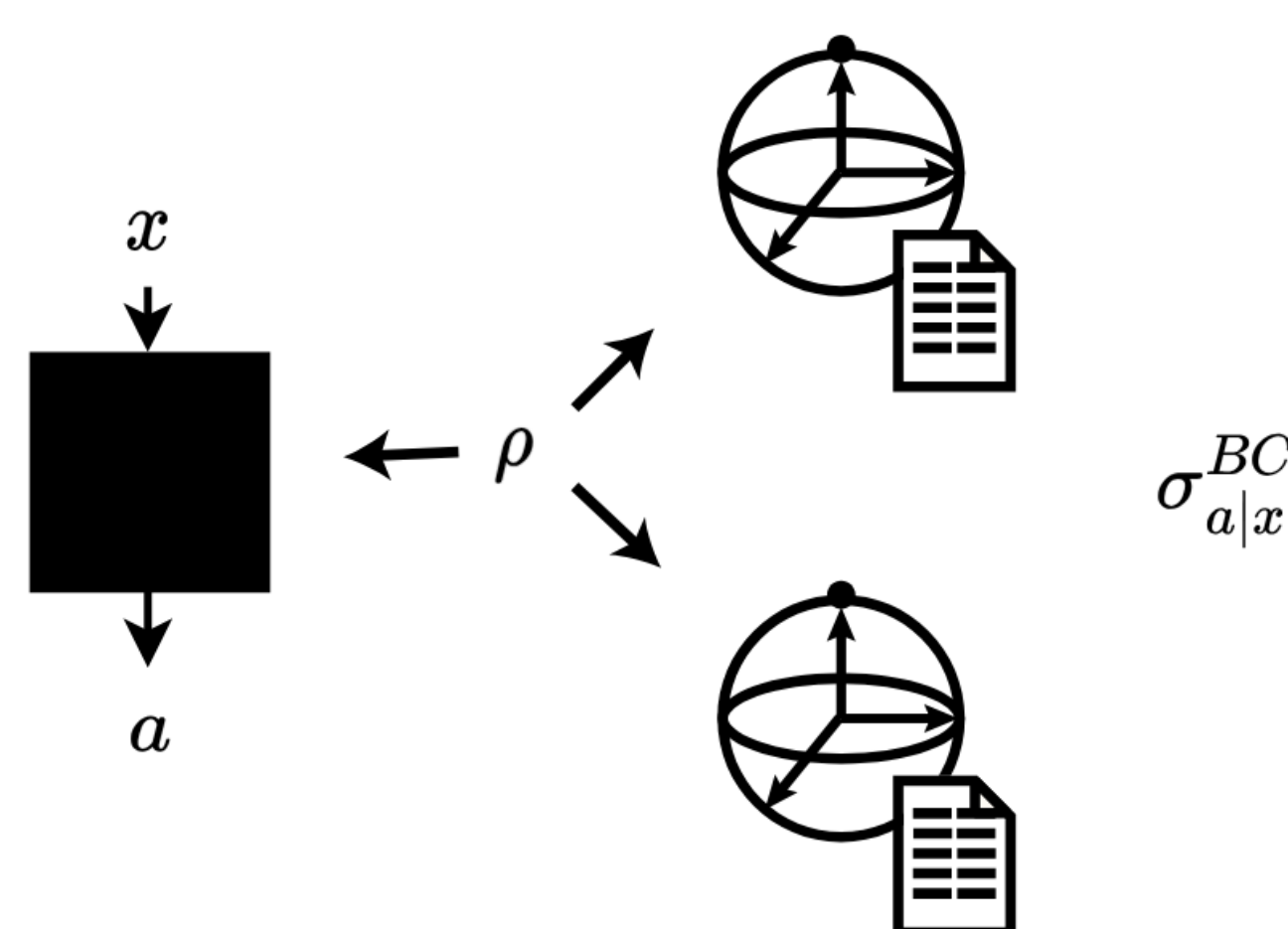


Figure 2: Scheme of a tripartite steering scenario with one uncharacterised party.

Steering assemblage: $\sigma_{a|x}^{BC} = \text{Tr}_A[(A_{a|x} \otimes \mathbb{1}^{BC}) \rho^{ABC}]$

Such assemblages do not demonstrate GMS when

$$\begin{aligned} \sigma_{a|x}^{BC} &= \sum_{\lambda} p(\lambda) p(a|x, \lambda) \rho_{\lambda}^{BC} + \sum_{\mu} p(\mu) \sigma_{a|x\mu}^B \otimes \rho_{\mu}^C \\ &+ \sum_{\nu} p(\nu) \rho_{\nu}^B \otimes \sigma_{a|x\nu}^C. \end{aligned}$$

This definition is motivated by one-sided device-independent certification of genuine multipartite entanglement.

Result 1

$\{A_{a|x}\}$ is incompatible $\Leftrightarrow \text{Tr}_A(A_{a|x} \otimes \mathbb{1}^{BC} |\text{GHZ}\rangle\langle\text{GHZ}|)$ exhibits GMS

Actually, any state of the form $|\psi\rangle = \sum_{i=0}^{d_A-1} \sqrt{p_i} |\phi_i^A\rangle \otimes |\zeta_i^{BC}\rangle$, where $p_i > 0$ and $|\zeta_i^{BC}\rangle$ is symmetric under the swap of B and C can be used! This includes, for example the W state, and some classes of Dicke states.

3.2. Two-sided device independent scenarios

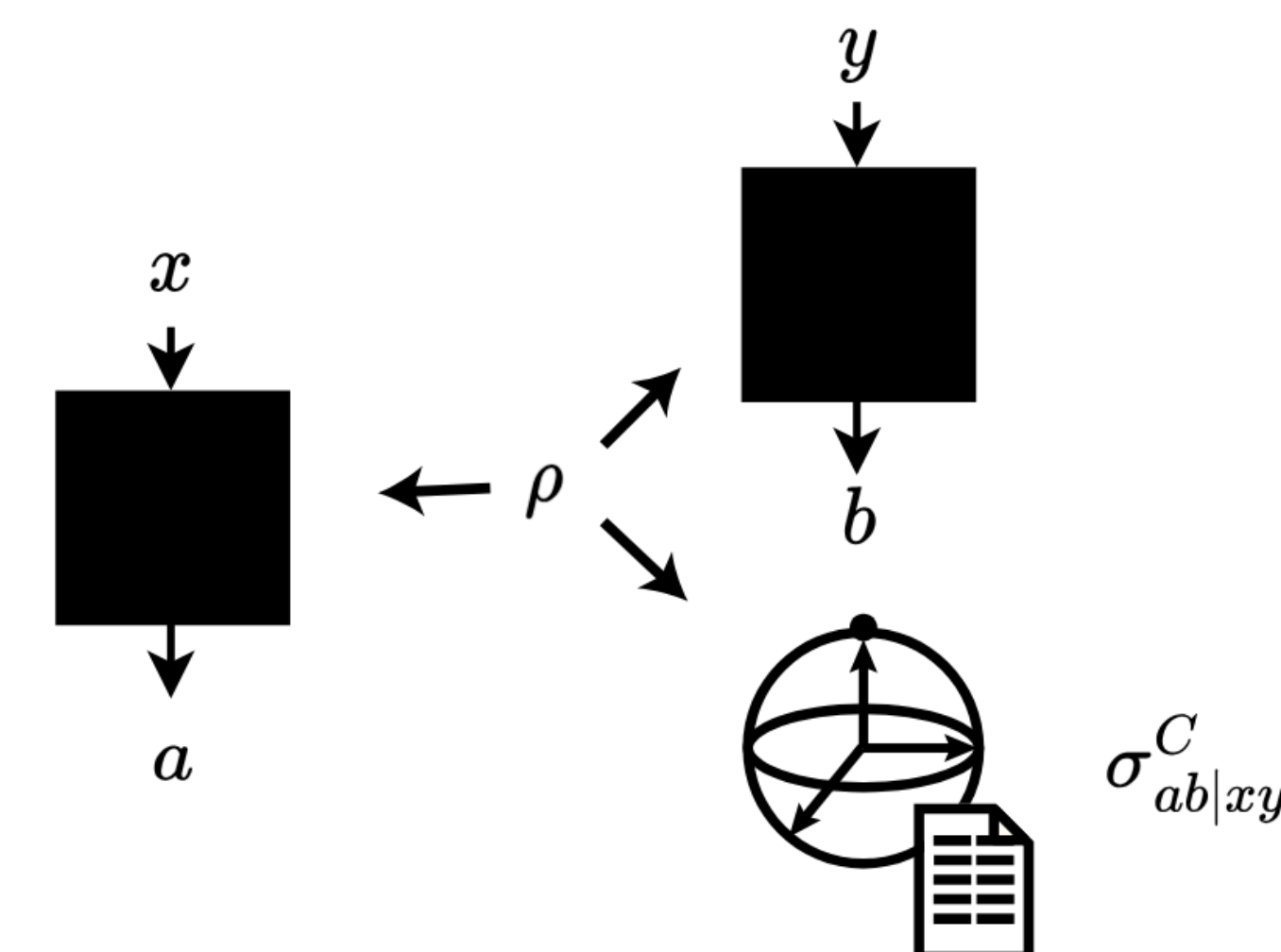


Figure 3: Scheme of a tripartite steering scenario with two uncharacterised parties.

Steering assemblage: $\sigma_{ab|xy}^C = \text{Tr}_{AB}[(A_{a|x} \otimes B_{b|y} \otimes \mathbb{1}^C) \rho^{ABC}]$

Such assemblages do not demonstrate GMS when

$$\begin{aligned} \sigma_{ab|xy}^C &= \sum_{\lambda} p(\lambda) p(a|x, \lambda) \sigma_{b|y\lambda}^C + \sum_{\mu} p(\mu) p(b|y, \mu) \sigma_{a|x\mu}^C \\ &+ \sum_{\nu} p(\nu) p(ab|xy, \nu) \rho_{\nu}^C. \end{aligned}$$

Result 2

Consider the incompatible set of measurements $\{M_{a|x}\}_{a,x}$ defined by

$$M_{\pm|x} = \frac{1}{2} \left(\mathbb{1} \pm \frac{1}{\sqrt{2}} \sigma_x \right),$$

where $x \in \{1, 2\}$ and σ_1, σ_2 are the Pauli matrices X and Y .

The assemblage $\text{Tr}_{AB}(M_{a|x} \otimes M_{b|y} \otimes \mathbb{1}^C \rho^{ABC})$ does not exhibit genuine multipartite steering for any ρ^{ABC} .

4. Discussions

- The close link between measurement incompatibility and bipartite steering extends only partially to the multipartite case.
- To prove Result 2, we introduce the decomposition

$$A_{a|x} \otimes B_{b|y} = \sum_{\lambda} p(a|x, \lambda) G_{\lambda, b|y} + \sum_{\mu} p(b|y, \mu) H_{\mu, a|x} + \sum_{\tau} p(ab|xy, \tau) F_{\tau},$$

which exactly characterises measurements that cannot lead to GMS in two-sided device-independent scenarios. This can be seen as a generalisation of joint measurability for spatially separated measurements, but a proper operational interpretation of such decomposition is still missing.

References

- [1] Marco Túlio Quintino, Tamás Vértesi, and Nicolas Brunner. Joint Measurability, Einstein-Podolsky-Rosen Steering, and Bell Nonlocality. *Phys. Rev. Lett.*, 113(16):160402, October 2014.
- [2] Roope Uola, Tobias Moroder, and Otfried Gühne. Joint Measurability of Generalized Measurements Implies Classicality. *Phys. Rev. Lett.*, 113(16):160403, October 2014.